

Chapter 9, Problem 4RQ

Problem

If $v_1 = 30 \sin(\omega t + 10^\circ)$ and $v_2 = 20 \sin(\omega t + 50^\circ)$, which of these statements are true?

- (a) v_1 leads v_2
- (b) v_2 leads v_1
- (c) v_2 lags v_1
- (d) v_1 lags v_2
- (e) v_1 and v_2 are in phase

Step-by-step solution

Step 1 of 3

Consider the voltage v_1 as,

$$v_1 = 30 \sin(\omega t + 10^\circ)$$

The phase of v_1 is,

$$\phi_1 = 10^\circ$$

Consider the voltage v_2 as,

$$v_2 = 20 \sin(\omega t + 50^\circ)$$

The phase of v_2 is,

$$\phi_2 = 50^\circ$$

Step 2 of 3

Compare the phases of v_1 and v_2 .

With respect to ϕ_1 , the phase of ϕ_1 is less than the phase of ϕ_2 .

Therefore, the voltage v_1 lags the voltage v_2 .

Therefore, the option (a) is wrong.

Step 3 of 3

With respect to ϕ_2 , the phase of ϕ_2 is greater than the phase of ϕ_1 .

Therefore, the voltage v_2 leads the voltage v_1 .

Therefore, the option (c), (e) are wrong.

Therefore, the correct options are $(b), (d)$.